

LABORATORY WORK 2

Normalization and DDL — International Airport Database

1. Unnormalized table

Booking_receipt(booking_id, passenger_full_name, passenger_passport_number, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, seat_numbers, ticket_price).

Normal form	Result	Reason
1NF	Violated	seat_numbers contains a comma-separated list.
2NF	Violated after 1NF repair	Passenger and flight attributes do not depend on the complete composite row key.
3NF	Violated after 1NF repair	Flight attributes create transitive dependencies through flight_number.

2. Normalization

2.1 First Normal Form (1NF)

One row is stored for each seat. Underlined values are primary-key attributes.

Table	Columns
Booking_receipt_1NF	booking_id, <u>passenger_passport_number</u> , passenger_full_name, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, <u>seat_number</u> , ticket_price

PK: (booking_id, passenger_passport_number).

2.2 Second Normal Form (2NF)

Table	Columns
Passenger_2NF	<u>passenger_passport_number</u> (PK), passenger_full_name
Booking_2NF	<u>booking_id</u> (PK), flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, ticket_price
BookingPassenger_2NF	<u>booking_id</u> (PK, FK), <u>passenger_passport_number</u> (PK, FK), seat_number

Partial dependencies are removed: passenger_full_name depends on passenger_passport_number; booking-level attributes depend on booking_id.

2.3 Third Normal Form (3NF)

Table	Columns
Passenger_3NF	<u>passenger_passport_number</u> (PK), passenger_full_name
Flight_3NF	<u>flight_number</u> (PK), departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name
Booking_3NF	<u>booking_id</u> (PK), flight_number (FK), ticket_price
BookingPassenger_3NF	<u>booking_id</u> (PK, FK), <u>passenger_passport_number</u> (PK, FK), seat_number

Transitive dependency booking_id → flight_number → airline_name is removed.

3. Functional dependencies

Dependency	Purpose
booking_id → flight_number, ticket_price	Booking attributes
passenger_passport_number → passenger_full_name	Passenger attributes
(booking_id, passenger_passport_number) → seat_number	Seat assignment
flight_number → departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name	Flight attributes
booking_id → flight_number → airline_name	Transitive dependency removed for 3NF

4. Flights table and 3NF

PK: flight_id.

Functional dependency: flight_id → all non-key attributes of Flights.

The table is in 3NF. departing_airport_id, arriving_airport_id, and airline_id are foreign keys. Airport and Airline descriptive attributes are stored in their own tables, so no transitive dependency exists inside Flights.

5. CREATE TABLE statements for 3NF tables

```
CREATE TABLE passenger_3nf (  
    passenger_passport_number VARCHAR(20) PRIMARY KEY,  
    passenger_full_name VARCHAR(100) NOT NULL  
);  
CREATE TABLE flight_3nf (  
    flight_number VARCHAR(20) PRIMARY KEY,  
    departure_airport_name VARCHAR(100) NOT NULL,  
    departure_city VARCHAR(50) NOT NULL,  
    arrival_airport_name VARCHAR(100) NOT NULL,  
    arrival_city VARCHAR(50) NOT NULL,  
    airline_name VARCHAR(100) NOT NULL  
);  
CREATE TABLE booking_3nf (  
    booking_id INT PRIMARY KEY,  
    flight_number VARCHAR(20) NOT NULL REFERENCES flight_3nf(flight_number),  
    ticket_price DECIMAL(7,2) NOT NULL  
);  
CREATE TABLE booking_passenger_3nf (  
    booking_id INT NOT NULL REFERENCES booking_3nf(booking_id),  
    passenger_passport_number VARCHAR(20) NOT NULL REFERENCES passenger_3nf(passenger_passport_number),  
    seat_number VARCHAR(10) NOT NULL,  
    PRIMARY KEY (booking_id, passenger_passport_number)  
);
```

6. INSERT statements

```
INSERT INTO passenger_3nf VALUES  
('P10001','Alice Johnson'),('P10002','Bob Smith'),('P10003','Carla Brown'),  
('P10004','David Wilson'),('P10005','Eva Davis');  
INSERT INTO flight_3nf VALUES  
('AF101','Heathrow Airport','London','Charles de Gaulle Airport','Paris','Air France'),  
('AF202','Charles de Gaulle Airport','Paris','Frankfurt Airport','Frankfurt','Air France'),  
('BA303','Heathrow Airport','London','JFK Airport','New York','British Airways'),  
('LH404','Frankfurt Airport','Frankfurt','Heathrow Airport','London','Lufthansa'),  
('EK505','Dubai International Airport','Dubai','Heathrow Airport','London','Emirates');  
INSERT INTO booking_3nf VALUES  
(1001,'AF101',220.00),(1002,'AF202',180.00),(1003,'BA303',650.00),  
(1004,'LH404',240.00),(1005,'EK505',720.00);  
INSERT INTO booking_passenger_3nf VALUES  
(1001,'P10001','12A'),(1002,'P10002','14B'),(1003,'P10003','18C'),  
(1004,'P10004','10D'),(1005,'P10005','22F');
```

7. DDL: required tables

```
CREATE TABLE Airline_info (  
    airline_id INT PRIMARY KEY, airline_code VARCHAR(30) NOT NULL,  
    airline_name VARCHAR(50) NOT NULL, airline_country VARCHAR(50) NOT NULL,  
    created_at TIMESTAMP NOT NULL, updated_at TIMESTAMP NOT NULL,  
    info VARCHAR(50)  
);  
CREATE TABLE Airport (  
    airport_id INT PRIMARY KEY, airport_name VARCHAR(50) NOT NULL,  
    country VARCHAR(50) NOT NULL, state VARCHAR(50) NOT NULL,  
    city VARCHAR(50) NOT NULL, created_at TIMESTAMP NOT NULL,  
    updated_at TIMESTAMP NOT NULL  
);  
CREATE TABLE Baggage_check (  
    baggage_check_id INT PRIMARY KEY, check_result VARCHAR(50) NOT NULL,  
    created_at TIMESTAMP NOT NULL, updated_at TIMESTAMP NOT NULL,  
    booking_id INT NOT NULL, passenger_id INT NOT NULL
```

```

);
CREATE TABLE Baggage (
    baggage_id INT PRIMARY KEY, weight_in_kg DECIMAL(4,2) NOT NULL,
    created_at TIMESTAMP NOT NULL, updated_at TIMESTAMP NOT NULL,
    booking_id INT NOT NULL
);
CREATE TABLE Boarding_pass (
    boarding_pass_id INT PRIMARY KEY, booking_id INT NOT NULL,
    seat VARCHAR(50) NOT NULL, boarding_time TIMESTAMP NOT NULL,
    created_at TIMESTAMP NOT NULL, updated_at TIMESTAMP NOT NULL
);
CREATE TABLE Booking_flight (
    booking_flight_id INT PRIMARY KEY, booking_id INT NOT NULL,
    flight_id INT NOT NULL, created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL
);
CREATE TABLE Booking (
    booking_id INT PRIMARY KEY, flight_id INT NOT NULL, passenger_id INT NOT NULL,
    booking_platform VARCHAR(50) NOT NULL, created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL, status VARCHAR(50) NOT NULL,
    price DECIMAL(7,2) NOT NULL
);
CREATE TABLE Flights (
    flight_id INT PRIMARY KEY, sch_departure_time TIMESTAMP NOT NULL,
    sch_arrival_time TIMESTAMP NOT NULL, departing_airport_id INT NOT NULL,
    arriving_airport_id INT NOT NULL, departing_gate TEXT NOT NULL,
    arriving_gate VARCHAR(50) NOT NULL, airline_id INT NOT NULL,
    act_departure_time TIMESTAMP, act_arrival_time TIMESTAMP,
    created_at TIMESTAMP NOT NULL, updated_at TIMESTAMP NOT NULL
);
CREATE TABLE Passengers (
    passenger_id INT PRIMARY KEY, first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL, date_of_birth DATE NOT NULL,
    gender VARCHAR(50) NOT NULL, country_of_citizenship VARCHAR(50) NOT NULL,
    country_of_residence VARCHAR(50) NOT NULL, passport_number VARCHAR(20) NOT NULL,
    created_at TIMESTAMP NOT NULL, updated_at TIMESTAMP NOT NULL
);
CREATE TABLE Security_check (
    security_check_id INT PRIMARY KEY, check_result VARCHAR(20) NOT NULL,
    created_at TIMESTAMP NOT NULL, updated_at TIMESTAMP NOT NULL,
    passenger_id INT NOT NULL
);

```

8. Required modifications

```

ALTER TABLE Airline_info RENAME TO Airline;
ALTER TABLE Booking RENAME COLUMN price TO ticket_price;
ALTER TABLE Flights ALTER COLUMN departing_gate TYPE TEXT;
ALTER TABLE Airline DROP COLUMN info;

```

9. Relationships

```

ALTER TABLE Security_check
ADD CONSTRAINT fk_security_passenger
FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id);

ALTER TABLE Booking
ADD CONSTRAINT fk_booking_passenger
FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id);

ALTER TABLE Baggage_check
ADD CONSTRAINT fk_baggage_check_passenger
FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id);

ALTER TABLE Baggage_check
ADD CONSTRAINT fk_baggage_check_booking
FOREIGN KEY (booking_id) REFERENCES Booking(booking_id);

ALTER TABLE Baggage
ADD CONSTRAINT fk_baggage_booking
FOREIGN KEY (booking_id) REFERENCES Booking(booking_id);

```

```
ALTER TABLE Boarding_pass
ADD CONSTRAINT fk_boarding_pass_booking
FOREIGN KEY (booking_id) REFERENCES Booking(booking_id);

ALTER TABLE Booking_flight
ADD CONSTRAINT fk_booking_flight_booking
FOREIGN KEY (booking_id) REFERENCES Booking(booking_id);

ALTER TABLE Booking_flight
ADD CONSTRAINT fk_booking_flight_flight
FOREIGN KEY (flight_id) REFERENCES Flights(flight_id);

ALTER TABLE Flights
ADD CONSTRAINT fk_flight_departing_airport
FOREIGN KEY (departing_airport_id) REFERENCES Airport(airport_id);

ALTER TABLE Flights
ADD CONSTRAINT fk_flight_arriving_airport
FOREIGN KEY (arriving_airport_id) REFERENCES Airport(airport_id);

ALTER TABLE Flights
ADD CONSTRAINT fk_flight_airline
FOREIGN KEY (airline_id) REFERENCES Airline(airline_id);
```